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Use of Stable Isotopes to Identify Redds of Putative Hatchery and Wild Atlantic Salmon and Evaluate Their Spawning Habitat and Egg Thiamine Status in a Lake Ontario Tributary

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Abstract

Both wild and hatchery Atlantic Salmon *Salmo salar* can contribute to restoration but can exhibit differences in spawning habitat selection (e.g., water depth, current speed) and egg nutritional quality (e.g., thiamine), which affect reproductive success. Hence, there is a need to be able to differentiate the spawning contribution of the two groups in the wild. As diets of wild and hatchery-reared spawners are markedly dissimilar and diet is known to influence stable isotope signature, egg stable isotope signatures offer the potential to discriminate redds of each. Using stable isotope analysis of carbon and nitrogen ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) of naturally spawned Atlantic Salmon eggs, we were able to discriminate the redds of putative wild (i.e., previously stocked life stage feeding in Lake Ontario; EWSR) from putative hatchery-reared Atlantic Salmon (EHSR). Eggs of EWSR were significantly more enriched in nitrogen ($\delta^{15}\text{N}$: $15.0 \pm 0.5\text{‰}$ [mean $\pm$ SE]) but more depleted in carbon ($\delta^{13}\text{C}$: $-26.6 \pm 0.3\text{‰}$) than eggs of EHSR ($\delta^{15}\text{N} = 9.8 \pm 0.6\text{‰}$; $\delta^{13}\text{C} = -17.5 \pm 0.0\text{‰}$). Eggs of EHSR were indistinguishable from eggs of known hatchery Atlantic Salmon for both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$. Using stable isotopes to discriminate redd type, few differences were found between the spawning habitat of putative wild and hatchery spawners. Similarly using the same criteria, thiamine levels in eggs of EWSR ($8,474 \pm 840$ pmol/g) were not significantly different from eggs of known wild Atlantic Salmon ($3,691 \pm 782$ pmol/g) or of eggs of EHSR ($14,865 \pm 1,050$ pmol/g), whose thiamine levels were indistinguishable from eggs of known hatchery Atlantic Salmon ($14,200 \pm 1,167$ pmol/g). Egg thiamine levels for all groups were above established mortality thresholds. Our results indicate that both hatchery and wild Atlantic Salmon can make reproductive contributions, which can be differentiated and quantified using stable isotope signatures.

Historically Lake Ontario may have supported the world's largest freshwater population of Atlantic Salmon *Salmo salar* (Webster 1982). By the late 1800s however, Atlantic Salmon were virtually extinct in the lake due to damming of tributaries that blocked spawning migration, overharvest, deforestation, and pollution (Netboy 1968; Parsons 1973; MacCrimmon

1977; Smith 1995). Attempts to reintroduce Atlantic Salmon to the Lake Ontario drainage through the 1900s had negligible success (Crawford 2001), possibly due in part to ongoing effects of an Alewife *Alosa pseudoharengus* diet-induced thiamine deficiency (Ketola et al. 2000) to which Atlantic Salmon reproduction may be very sensitive (Fisher et al. 1996).

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Alewives contain high levels of thiaminase activity, which is presumed to result in thiamine destruction in the gut of a predator as a postmortem change (Tillitt et al. 2005). As a result, a diet dominated by Alewives can lead to thiamine deficiency. In Great Lakes salmonines other than Atlantic Salmon, diets dominated by Alewives are associated with low egg thiamine (Fitzsimons and Brown 1998; Honeyfield et al. 2005), while the thiamine content of Atlantic Salmon eggs from the Finger Lakes (Fisher et al. 1998a, 1998b) and Baltic Sea (Amcoff et al. 1999; Börjeson et al. 1999) has been related to the thiamine nutrition of the maternal parent. The salmonine egg receives all of its thiamine, an essential nutrient, from the maternal parent (Koski et al. 2005), such that the thiamine content of a salmonine egg will reflect the diet of the maternal parent. The thiamine content of the diet of hatchery-reared salmonids tends to be much higher than that of diets of wild salmonids, resulting in much higher egg thiamine concentrations (Fitzsimons et al. 1998; Fisher et al. 1998b). Because of these differences, Fisher et al. (1998a) found that thiamine levels in Atlantic Salmon fed a hatchery diet were five-fold higher than those feeding on Rainbow Smelt *Osmerus mordax* and from approximately 8- to 80-fold higher than those feeding on Alewife. The problems of maternal thiamine nutrition associated with Alewife consumption may be further exacerbated in Atlantic Salmon because they become anorexic before entering the spawning river (Kadri et al. 1995), which may occur several months prior to spawning (J. Kendell, Credit River Anglers Association, personal communication), and do not feed during the several months they spend in the river (Grey and Tosh 1894 cited in Kadri et al. 1995). The prolonged period of reduced thiamine intake during this time would worsen a preexisting thiamine deficiency causing lower than expected thiamine levels in eggs leading to higher than expected embryonic mortality (Karlsson et al. 1999).

As a result of changes in the Lake Ontario food chain beginning in the late 1990s (Mills et al. 2003), Alewife underwent a dramatic decline in abundance, thus reducing the potential for thiamine deficiency (Honeyfield et al. 2005; Fitzsimons et al. 2009, 2010). In the early 2000s, there was evidence of natural reproduction by salmonines other than Atlantic Salmon in tributaries throughout Lake Ontario suggesting improvement in habitat quality (McKenna and Johnson 2005; Connerton et al. 2009) and that thiamine levels, for at least some species, were adequate for reproduction. As a result of these changes, management agencies took actions involving the stocking of multiple life stages of Atlantic Salmon, including adults, and the assessment of bottlenecks to determine if conditions for successful salmonid reproduction had improved from those existing earlier. Despite these efforts there remains considerable uncertainty regarding the basic biology of Atlantic Salmon, including the diet of adults while in Lake Ontario.

The spawning habitat used by Atlantic Salmon has been widely studied, but there remain concerns about spawning habitat selection by hatchery-reared spawners relative to that of wild spawners. Based on laboratory experiments, hatchery-reared

Atlantic Salmon may select inferior habitat for spawning relative to their wild counterparts (Fleming et al. 1996), and this can have negative implications for offspring survival and growth (Magee et al. 1996; Bernier-Borgault and Magnan 2002). The spawning habitat used by wild Atlantic Salmon has been described for several rivers in Europe (Moir et al. 1998; Heggberget et al. 1988) and eastern North America (Gibson 1993) but not for tributaries of Lake Ontario. Scott et al. (2005) described the spawning habitat characteristics used by hatchery-reared Atlantic Salmon in one Lake Ontario tributary that were found to differ from the characteristics of random locations within the tributary. Initially, because there is no wild remnant stock, it is expected most spawning by Atlantic Salmon in Lake Ontario tributaries will be the result of hatchery-reared fish with increasing contributions from fish resulting from natural reproduction over time but initially dependent on the reproductive success of hatchery-reared fish.

Differentiating redds of wild and hatchery-reared Atlantic Salmon and evaluation of their respective spawning habitat and egg thiamine content is essential for understanding linkages with embryo survival. As wild and hatchery-reared Atlantic Salmon spawn at the same time, separation of redds on a temporal basis is not possible (Heggberget et al. 1988). Separation of juvenile and adult wild and hatchery-reared fish has been carried out using scale pattern analysis (Hansen et al. 1993), morphological characteristics (Crozier 1998), or a combination of these methods (Carr et al. 1997). Occasionally use of carotenoid pigment analyses have been applied (Youngson et al. 1997). In general however, no methods are totally discriminatory and a combination of methods is often recommended, but few are appropriate for eggs (Lund and Hansen 1991). In addition, many of these methods require sampling of adults, which for Atlantic Salmon is difficult for a number of reasons. Prior to spawning, adult Atlantic Salmon are often cryptic, seeking cover in deep pools or overhanging banks (Armstrong et al. 2003; J. D. Fitzsimons, unpublished data), and most likely at low abundance (Carr et al. 2004). In addition, sampling of adults prior to spawning may alter their spawning behavior and so affect redd characteristics (Thorstad et al. 2003). Moreover, once Atlantic Salmon have spawned, adults generally leave the spawning area (Baglinière et al. 1990) making it difficult to associate adults with their corresponding redds.

An alternative technique that could identify spawning by hatchery fish from that of wild fish postspawning is the analysis of spawned eggs for stable isotope signatures ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$). Stable isotopes have been used to infer movement patterns of fishes (Hesslein et al. 1991), distinguish anadromous from nonanadromous components of a stock (Doucett et al. 1999a, 1999b), identify different spawning stocks or populations of fishes (Gao and Beamish 1999), and provide information about feeding relationships in aquatic environments (Guiguer et al. 2002). Stable isotopes were used to differentiate farmed Atlantic Salmon fed on a homogenous commercial aquaculture diet, from wild Atlantic Salmon, presumed to be feeding on a

range of wild prey (Dempson and Power 2004). Stable isotopes yield an assimilated history of feeding relationships (Fry and Sherr 1984; Hobson and Welch 1992), which for freshwater Atlantic Salmon probably reflects lacustrine feeding because they become anorexic prior to the spawning migration (Kadri et al. 1995).

We hypothesize that eggs spawned by hatchery-reared Atlantic Salmon, fed a commercial hatchery diet, would yield characteristically different stable isotope signatures than eggs of Atlantic Salmon that feed on wild prey in Lake Ontario and that had been stocked previously, being defined here as wild Atlantic Salmon. Further we hypothesize that if eggs can be differentiated on the basis of stable isotope signature, this information can be used to (1) determine whether captive breeding and rearing of Atlantic Salmon in a hatchery environment affects spawning site selection relative to wild spawners and (2) determine whether the thiamine content of spawners using a wild diet containing unknown amounts of Alewives, which cause thiamine deficiency, differs from spawners fed a hatchery diet assumed to have adequate thiamine levels (Fisher et al. 1998b).

The aim of the present study was to examine the utility of stable isotope analyses to discriminate redds of hatchery-reared and wild Atlantic Salmon in the Credit River and their respective habitat attributes, egg thiamine content, and adequacy of thiamine levels for reproduction. Redds were constructed by a combination of wild Atlantic Salmon that had migrated upstream to the study area from Lake Ontario and hatchery-reared Atlantic Salmon that were stocked as spawning adults directly into the study area.

METHODS

Atlantic Salmon stocking: hatchery-reared fish.—Fish used for stocking in the stocking area located on the upper Credit River approximately 40 km upstream of Lake Ontario (Figure 1) were captive bred age-4+ sexually mature adult LeHavre strain Atlantic Salmon. They had been reared at the Ontario Ministry of Natural Resources Fish Culture Station (OMNRFCS) in Harwood, Ontario. The lineage of the fish used in this study is one that has been subject to artificial breeding and captivity for the past 18 years (about 3–5 generations; Normandale OMNRFCS) and artificial breeding since 1969 (P. Amiro, Fisheries and Oceans Canada, personal communication). Stocking occurred at three locations within a 1,500-m section of the upper Credit River (Ontario) approximately 40 km upstream of its mouth on Lake Ontario. Stocking occurred on October 14, 2009, just prior to the 2009 spawning season. Each of the three stocking locations corresponded to road crossings (Figure 2), and at each location 114 Atlantic Salmon were released. Based on detailed hatchery records on gender and maturity, which we assumed applied to the Atlantic Salmon released, we assumed that of the 114 fish released at each location, 48% (54) were mature females, 48% (54) were mature males, and 4% (6) were of unknown gender and unlikely to spawn (D. Rosborough, Ontario

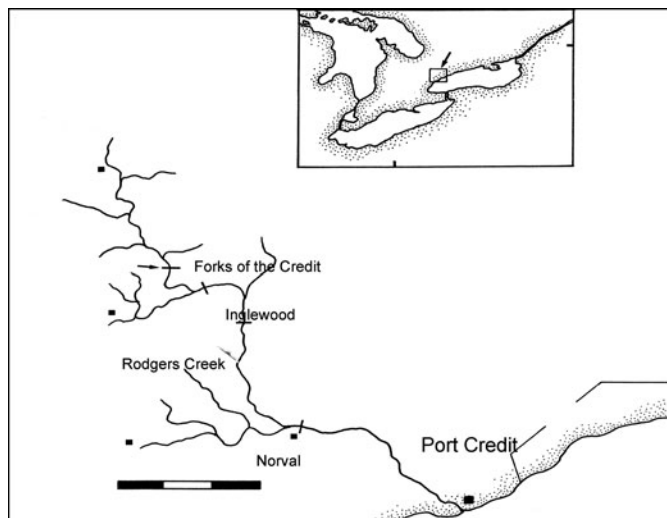


FIGURE 1. Map of the location of the Credit River (Ontario), a tributary of Lake Ontario, located within the Great Lakes basin. Measurement bar indicates 15 kilometers.

Ministry of Natural Resources, personal communication). Historically Atlantic Salmon used the upper Credit River for spawning, and earlier work (Fitzsimons, unpublished data) involving stocked hatchery-reared Atlantic Salmon indicated that of several areas where they were stocked in equal numbers and sex ratios, the highest spawning per adult female occurred in this area. Although Brown Trout *Salmo trutta* are also present in the upper Credit River and spawning habitat characteristics of Brown Trout are similar to those of Atlantic Salmon (Heggberget et al. 1988; Armstrong et al. 2003), the area where Atlantic Salmon were released was upstream of the area reported used by resident Brown Trout for spawning (Zimmer and Power 2006). At each stocking location, individual Atlantic Salmon were dipnetted from a 92,270-L tank that was used for their transport from the OMNRFCS. Atlantic Salmon were placed directly into a pool in the river with most fish dispersing either upstream or downstream within hours of stocking. No wild Atlantic Salmon were stocked in the area where hatchery Atlantic Salmon were released and thus it was assumed that any wild Atlantic Salmon present had migrated there from Lake Ontario although at the time, no runs of wild Atlantic Salmon were known to exist.

Atlantic Salmon stocking: wild fish.—To evaluate redd habitat selection and construction by wild Atlantic Salmon in the absence of hatchery-reared Atlantic Salmon, wild Atlantic Salmon from Lake Ontario were stocked into Rodgers Creek (Figure 1), a small tributary of the Credit River. Wild Atlantic Salmon for stocking into Rodgers Creek were collected from the Credit River at Streetsville on their upstream migration from Lake Ontario. Streetsville is located approximately 10 km upstream of Lake Ontario, while Rodgers Creek is approximately 30 km upstream. Atlantic Salmon were collected by angling or electrofishing below the Streetsville fishway or from the fishway itself between August 28 and November 2, 2009. On the day of capture, fish

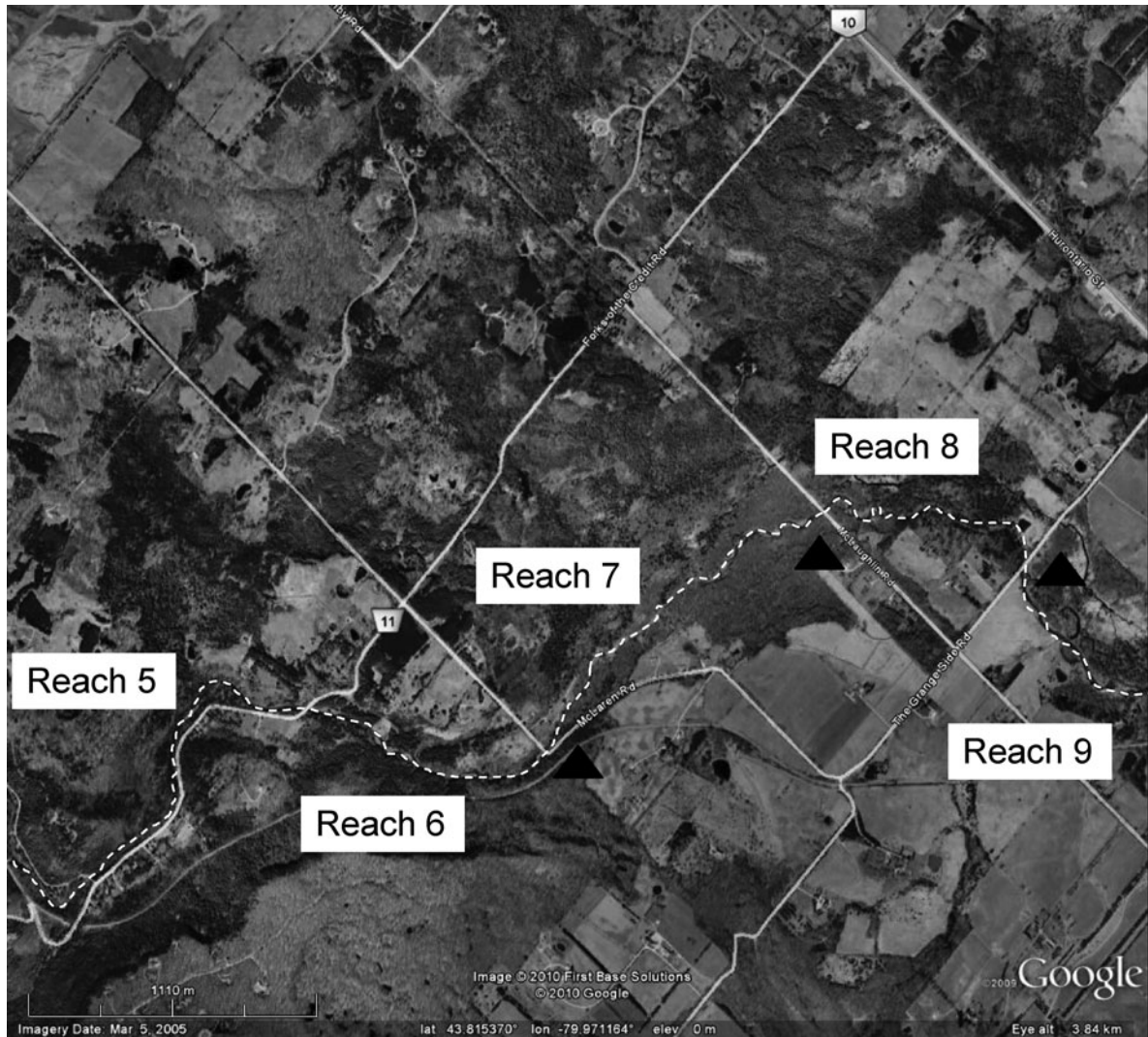


FIGURE 2. Locations (black triangles) where Atlantic Salmon stocking took place on the Credit River in 2009, and the five reaches (reach 5 is upstream, reach 9 is downstream) evaluated for evidence of spawning activity and used for the collection of eggs from the redds. Road crossings demarcate the upstream and downstream extent of the reaches. The dotted line denotes the approximate river course.

were measured for length (mm) and sexed using external morphology before being moved by road in a splash tank mounted on a truck and stocked into a pool just below a small dam on Rodgers Creek. No hatchery-reared Atlantic Salmon were stocked into Rodgers Creek and because of obstructions near its confluence with the Credit River, it was considered unlikely that Atlantic Salmon would enter Rodgers Creek of their own volition from the Credit River.

Redd surveys and redd excavation.—To identify and enumerate Atlantic Salmon redds on the upper Credit River, a total of four relatively separate reaches (Figure 1) were traversed weekly starting October 1, three weeks after Atlantic Salmon had been stocked. Redd surveys were also conducted on Rodgers Creek. All redd locations were identified in comparison to surrounding algal- or sediment-covered substratum as areas of clean

substratum with characteristic structure (Witzel and MacCrimmon 1983) and further confirmed as putative redds through the identification of upstream pits and downstream tailspills (Schmetterling 2000). Global Positioning System (GPS) coordinates were taken for all redds found, and the downstream extent of the redd was marked by inserting a colored peg securely into the substrate.

For each redd we measured flow rate (cm/s), water depth (cm), egg pocket depth (cm), redd area (m²), stream width (m), canopy (%), riparian vegetation type and cover, and stream habitat type (e.g., riffle, pool–riffle, run). Water depth was measured at the center of the redd. Egg pocket depth was defined as the downward vertical extent of loose gravel at the center of the redd. We did not make measures of substrate size as it has already been extensively reported that Atlantic Salmon use gravel-size

substrate for spawning (Heggberget et al. 1988; Armstrong et al. 2003; Scott et al. 2005) and hatchery and wild fish appear to use similar habitat for spawning (Økland et al. 1995; Fleming et al. 1997).

As a basis for determining whether Atlantic Salmon spawning habitat selection was deliberate or random in nature, we evaluated habitat at redd and nonredd sites on the Credit River. To identify nonredd random sites we used a table of random numbers to produce a set of coordinates for random sites within the study area. We compared the habitat characteristics for these random sites with that of redd sites. Occasionally coordinates for random sites would require sampling a location that would not be possible for a fish to use for spawning, such as too close to the shore or under a fallen tree, or that was not possible for us to sample, such as in a deep pool. In such cases, we did not collect data from the location. Exclusion of such locations is reasonable because it is not representative to include unusable spawning sites in an assessment of site selection.

As spawners can construct test pits that superficially resemble redds but do not contain eggs, it was necessary to verify that redds contained eggs; for verification we excavated each redd. As eggs immediately after spawning and for some time after that exhibit elevated mortality, possibly as a result of sensitivity to physical shock (Jensen and Alderdice 1989), we delayed excavating redds until March, by which time eggs had reached the eyed stage.

Redds were excavated with a flat shovel (20 cm long) while two overlapping collection nets (25-cm diameter; 800–900- μ m multifilament mesh) were held in the current immediately downstream of the redd. While facing the stream bank, the shovel was inserted vertically up to a depth of at least 20 cm into the substrate, and hence deeper than the average egg pocket depth (15 cm) reported for Atlantic Salmon (Heggberget et al. 1988) and other salmonines possibly involved in redd construction (Zimmer and Power 2006; Devries 1997). The shovel blade was then moved through a 90° arc while being lifted vertically until horizontal. The shovel blade and contents, while held in a horizontal orientation, were lifted to midway in the water column where the shovel blade was gently rocked in the water current to dislodge fine material and eggs into one of two downstream collection nets. This process was repeated until either eggs were collected or 75% of the length of redd had been excavated and no eggs found. For Rodgers Creek, because of concerns about egg mortality resulting from the redd excavation procedure, only 25% of the redd length was excavated. All eggs from an individual redd were placed into an individual 250-mL plastic bottle filled with river water and examined later the same day. When examined, eggs were separated into translucent-live, or opaque-dead egg categories. Only live eggs were used for stable isotope, thiamine, and genetic analysis (see below).

Collection of eggs from migratory wild Atlantic Salmon.—Known wild Atlantic Salmon collected by angling or electrofishing below the Streetsville fishway that were initially intended for stocking into Rodgers Creek (see above) but which died during capture or shortly thereafter in transit in river water were

used to provide samples for stable isotope and thiamine analysis. Because of death these samples were potentially biased, but stable isotopes are robust to a variety of preservation methods (Barrow et al. 2008) and salmonine eggs have been held for periods of several months in a hatchery with no change in thiamine content (Brown et al. 1998a), suggesting that the maximum 1-h holding would have had minimal effects. Moreover, thiamine levels in this group of eggs were all above those of Atlantic Salmon that migrated successfully to a collection weir on the St. Mary's River (Michigan) from Lake Huron, a distance of 120 km (Fitzsimons, unpublished data). As there were no barriers on the Credit River downstream of this point and the location is only 10 km from Lake Ontario, Atlantic Salmon collected at this location were considered representative of the Atlantic Salmon in the lake. Dead Atlantic Salmon were immediately placed on ice and transferred to a freezer (−20°C) for storage prior to eggs being removed for stable isotope and thiamine analysis. Frozen fish were measured for total length (mm) and weight (g).

Stable isotope analysis.—Egg samples for stable isotope analysis of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ from either redds or females were dried to a constant mass at 65°C and ground to a fine powder. Lipids were extracted from all tissues before analysis regardless of isotope (see methods in Folch et al. 1957; Post and Parkinson 2001) to account for the differential routing of stable isotopes in adipose versus dorsal muscle tissue (Kling et al. 1992). Although lipid extraction can significantly increase $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ causing a positive shift in overall food web placement, the effects do not appear to be species-specific such that overall food web placement in Lake Ontario should not have been altered (Murry et al. 2006). Stable isotope measurements were performed using a Finnigan MAT Delta Plus continuous-flow elemental analyzer at Cornell University's Boyce Thompson Stable Isotope Facility (Ithaca, New York). Samples with high nitrogen content were compared with a methionine standard ($\delta^{13}\text{C} = -25.40\text{‰}$, $\delta^{15}\text{N} = -0.81\text{‰}$, 40.25% C, 9.39% N) and samples with low nitrogen content were compared with a burned cabbage standard ($\delta^{13}\text{C} = -27.22\text{‰}$, $\delta^{15}\text{N} = 0.77\text{‰}$, 41.60% C, 3.35% N). The standard error from the mean of each isotopic run never exceeded 0.15‰ with respect to the standards. Stable isotope ratios are expressed in δ notation as the deviation from standards in ppt (‰) according to the following equation: $\delta^{13}\text{C}$, $\delta^{15}\text{N} = (\text{R}_{\text{sample}}/\text{R}_{\text{standard}} - 1) \times 10^3$, where $\text{R} = {}^{13}\text{C}/{}^{12}\text{C}$ or ${}^{15}\text{N}/{}^{14}\text{N}$. The standard reference material was atmospheric nitrogen for ${}^{15}\text{N}$ and Pee Dee Belemnite carbonate for ${}^{13}\text{C}$.

Thiamine analysis.—Thiamine analysis of eggs from redds or females was by the high pressure liquid chromatography method of Brown et al. (1998b). The individual thiamine vitamers thiamine pyrophosphate, thiamine monophosphate, and free thiamine were quantified and thiamine concentration reported as the sum of all three vitamers.

Hatchery Atlantic Salmon.—To serve as a reference for thiamine and stable isotopes for hatchery Atlantic Salmon eggs collected from redds, eggs of the same strain (LeHavre) held at a hatchery and fed the same diet as the Atlantic Salmon stocked

(see above) were sampled for stable isotopes and thiamine analysis (hereafter hatchery females).

Genetic analysis.—Species identification and confirmation that redds were built by Atlantic Salmon and not Brown Trout, which are also known to spawn in the Credit River (Zimmer and Power 2006), was accomplished by genetic species identification of eggs collected from each redd. The DNA from individual eggs was extracted, amplified, and sequenced for a 500-nucleotide portion of the mitochondrial cytochrome b (cyt-b) gene, using methods described in Kyle and Wilson (2007). Species identification was confirmed using a local BLAST search (Altschul et al. 1997) against species-diagnostic reference sequences in a wildlife forensic database (Kyle and Wilson 2007). The DNA sequences from egg samples were tested for homology (identity) with reference sequences from Atlantic Salmon, Brown Trout, and other salmonids (Kyle and Wilson 2007); egg sequences with >99% homology with species-diagnostic reference sequences were recognized as belonging to that species.

Statistics.—We used *t*-tests to compare the habitat criteria of redds to those of random samples and to compare the habitat criteria of wild and hatchery Atlantic Salmon. Then *G*-tests were used to assess the independence between the type of habitat (e.g., run, riffle) where redds were constructed and either the spawner type (e.g., wild, hatchery) or the habitat category (e.g., random location, redd). Comparison of stable isotopes and thiamine to assess the effect of putative parentage (e.g., hatchery, wild) and source (e.g., redd, female) of eggs was done with one-way analysis of variance using the general linear model procedure in Systat. To conform to assumptions of normality and homoscedasticity, all data were $\log_{10}$ transformed prior to analysis and the appropriateness of this transformation confirmed with normal probability plots. All tests were declared significant at $P < 0.05$.

RESULTS

Comparison of Hatchery and Wild Atlantic Salmon Spawning Activity and Stable Isotopes

A total of 79 potential Atlantic Salmon redds were identified in the reaches of the upper Credit River. These were associated

with an unknown number of wild spawners and the 342 hatchery Atlantic Salmon stocked into the upper Credit River during 2009 (Figure 2). Of the 79 suspected redds, 34 were excavated and found to contain eggs, all of which were confirmed by genetic analysis to be those of Atlantic Salmon. A much smaller number of wild Atlantic Salmon (18) were stocked into Rodgers Creek consisting of 10 males and 8 females having an average total length of 531 mm. On Rodgers Creek, a total of 23 potential redds were identified although eggs could only be extracted from five of these redds and all of these were confirmed as being those of Atlantic Salmon.

Due to limited numbers of eggs obtained from individual redds on the upper Credit River, eggs from only 20 redds from the Credit River were analyzed for stable isotopes, and these were compared to eggs obtained from the five known wild spawners collected from the fishway at Streetsville and four broodstock fish from the hatchery. Insufficient sample material was available from redds from Rodgers Creek for meaningful statistical analysis. Results for the Credit River for both $\delta^{13}\text{C}$ ($F = 860.1$; $df = 3, 24$; $P < 0.001$) and $\delta^{15}\text{N}$ ($F = 266.1$; $df = 3, 24$; $P < 0.001$) indicated a significant effect of the type of egg sample (redds or females) and their putative parentage (hatchery or wild) (Table 1). The biplot of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ for eggs sampled from redds on the Credit River (Figure 3) showed clear separation of one group of redds, presumed to be constructed by wild spawners, from another group of redds, presumed to be constructed by hatchery spawners, and this was evident for both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ (Figure 3). This pattern was generally evident across the four reaches examined although insufficient numbers of eggs were collected to examine whether wild and hatchery spawners preferred different reaches. The presumption that dissimilar isotopic signatures represented differences in parentage (wild versus hatchery) was further supported by the following evidence. Stable isotope signatures differed significantly between putative wild and hatchery types. For eggs collected from known wild spawners at Streetsville, $\delta^{15}\text{N}$ (16.4‰) was significantly higher than that of putative wild redd eggs (15.0‰) collected from the upper Credit River, and much higher than that of eggs from known hatchery fish (9.9‰) or eggs from redds of putative hatchery fish (9.8‰). Similarly for putative wild redds, although $\delta^{13}\text{C}$ (−26.6‰) was significantly less than that of known wild spawners (−22.2‰), it was much less than that

TABLE 1. Summary of egg thiamine for wild and hatchery Atlantic Salmon eggs removed from females or redds on the Credit River and Rodgers Creek in 2009. Means followed by the same lowercase letter in the same column are not significantly different ($P > 0.05$).

Source of eggs	Type of female or redd	Location	N	Thiamine mean (SE)
Females	Wild, known	Credit River	6	3,692 (783) y
Redds	Wild, putative	Credit River	14	8,474 (840) zy
Redds	Hatchery, putative	Credit River	6	14,865 (1,050) z
Redds	Wild, known	Rodgers Creek	5	6,511 (1,931) y
Female	Hatchery, known	Hatchery	5	14,200 (1,167) z

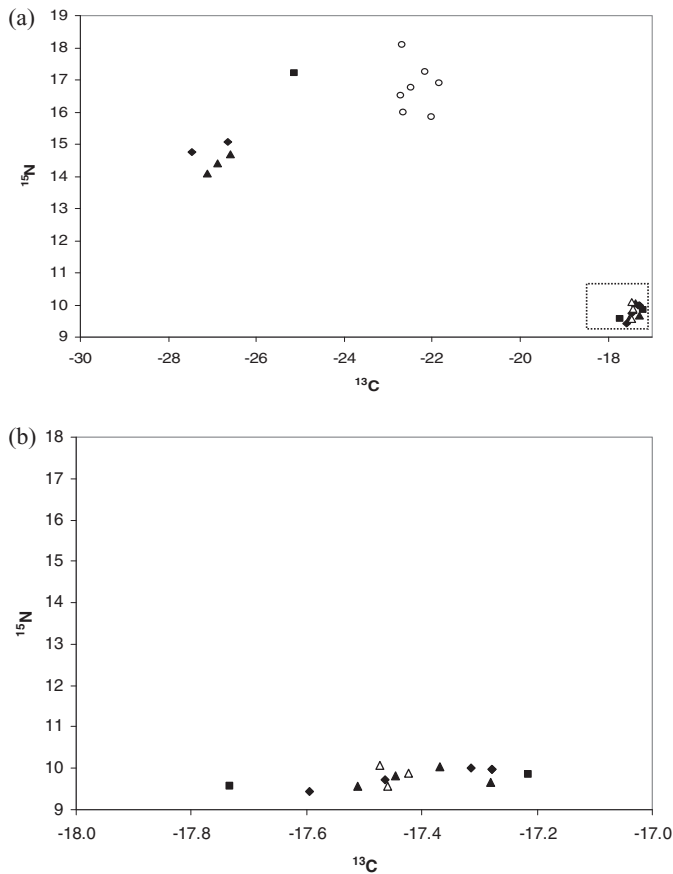


FIGURE 3. Biplot of $\delta^{15}\text{N}$ (‰) and $\delta^{13}\text{C}$ (‰) for Atlantic Salmon eggs removed from redds from four reaches of the Credit River, Rodgers Creek, or wild Atlantic Salmon females for (a) all data and (b) a restricted set of $\delta^{13}\text{C}$ (to highlight eggs of hatchery parents; shown as a box in the lower right of Figure 3a). Symbols are as follows: filled diamond = reach 6, filled triangle = reach 7, open triangle = reach 8, filled square = reach 9, open diamond = adults.

of eggs of known hatchery fish (-17.7‰) or eggs from redds of putative hatchery fish (-17.5‰). The differences among redd types are unlikely to be the result of different sized spawners constructing redds as neither $\delta^{13}\text{C}$ nor $\delta^{15}\text{N}$ in eggs of wild spawners was related ($P > 0.05$) to the size of parent whose total length ranged from 551 to 607 mm.

Comparison of Hatchery and Wild Atlantic Salmon Spawning Habitat Selection

Using the stable isotope data to assign putative redd parentage (e.g., wild or hatchery), the spawning habitat used on the upper Credit River by hatchery spawners was largely indistinguishable from that of wild spawners with two exceptions. These exceptions were the type of habitat Atlantic Salmon spawned in and the proportion of riparian vegetation associated with the spawning area. The type of habitat used for spawning (e.g., riffle, pool-riffle, run) differed ($G = 15.3$, $df = 2$, $P < 0.05$) between wild and hatchery spawners; wild Atlantic Salmon appeared to prefer pool-riffle (60%) over riffle (40%) or run (0%)

TABLE 2. Summary of habitat observations for hatchery and wild Atlantic Salmon spawners on the Credit River in 2009. An asterisk indicates $P < 0.05$.

Parameter	Hatchery redd mean (SE)	Wild redd mean (SE)	T , P -value
N	12	6	
Canopy (%)	5.9 (2.8)	9.0 (4.0)	-0.6, 0.27
Riparian vegetation (%)	58.6 (5.1)	86.0 (9.3)	-2.6, 0.02*
Stream width (m)	14.8 (1.0)	14.9 (2.0)	-0.1, 0.47
Distance from nearest shore (m)	3.8 (0.7)	4.3 (0.9)	-1.4, 0.34
Water flow (m/s)	0.6 (0.0)	0.7 (0.1)	-1.4, 0.12
Water depth (cm)	38.6 (4.4)	28.4 (8.0)	-1.4, 0.10
Pit depth (cm)	9.8 (0.4)	10.9 (0.9)	-1.2, 0.14
Redd width (m)	0.8 (0.1)	1.0 (0.2)	-1.4, 0.12
Redd length (m)	1.5 (0.2)	2.0 (0.3)	-1.4, 0.10
Redd area (m^2)	1.1 (0.2)	1.9 (0.8)	-1.10, 0.16
Spawning days after November 1	10.7	12.2	

habitat, whereas hatchery-reared Atlantic Salmon preferred riffle (64%) over pool-riffle (27%) or run (9%) habitat. Wild Atlantic Salmon spawned in areas with significantly greater amounts of riparian vegetation (86%) than hatchery-reared (59%) fish (Table 2). The limited data available for wild redds on Rodgers Creek (Table 3) indicated that wild Atlantic Salmon selected similar habitat on this smaller system with the exception of stream width, which was as expected less (4.0 m) than for the larger upper Credit River (14.8 m), and distance from the nearest shore (1.8 m), which was less than one half that of the Credit River (3.8 m).

TABLE 3. Summary of habitat observations for Atlantic Salmon redds on Rodgers Creek in 2009.

Parameter	Wild redd mean (SE)
N	5
Canopy (%)	16.0 (4.0)
Riparian vegetation (%)	83.0 (7.7)
Stream width (m)	4.0 (0.3)
Distance from nearest shore (m)	1.3 (0.3)
Water flow (m/s)	0.4 (0.0)
Water depth (cm)	13.2 (1.5)
Pit depth (cm)	8.6 (1.0)
Redd width (m)	0.6 (0.1)
Redd length (m)	1.3 (0.2)
Redd area (m^2)	0.9 (0.3)
Spawning days after November 1	13.2

TABLE 4. Summary of habitat observations for all Atlantic Salmon redds (wild, hatchery) and random locations on the Credit River in 2009. An asterisk indicates a significant difference ($P < 0.05$) between groups.

Parameter	All redds mean (SE)	Random locations mean (SE)	T , P -value
N	18	41	
Canopy (%)	6.9 (2.2)	10.6 (2.5)	−1.1, 0.13
Riparian vegetation (%)	57.2 (5.4)	67.0 (3.4)	0.3, 0.50
Stream width (m)	14.8 (0.9)	11.6 (0.8)	2.7, 0.005*
Water flow (m/s)	0.6 (0.0)	0.4 (0.1)	3.8, 0.0002*
Water depth (cm)	35.4 (4.0)	50.7 (4.4)	−2.6, 0.007*

Spawning fish, whether putative wild or hatchery in origin, selected spatial habitat in proportion significantly different from available habitat (Table 4). The width of stream used by spawning Atlantic Salmon was wider (14.8 m) than was available (11.6 m), with spawning occurring in shallower water depths (35.4 cm) than were available (50.7 cm) but at higher flows (0.6 m/s) than were available (0.4 m/s). The proportion of type of habitat used for spawning (e.g., riffle, pool–riffle, run, pool) did not differ from the proportions available ($G = 7.5$, $df = 3$, $P > 0.05$).

Comparison of Hatchery and Wild Atlantic Salmon Thiamine Status

For the upper Credit River, when stable isotope data were used to assign putative redd identity (e.g., wild or hatchery), egg thiamine status varied ($F = 13.2$; $df = 4, 28$; $P < 0.001$) depending on the source of eggs (redd versus female) but not whether redds were of wild or hatchery origin (Table 1). Overall the highest egg thiamine concentration occurred in eggs from putative hatchery redds (14,865 pmol/g), but they were not different from eggs of known hatchery fish (14,200 pmol/g). Similarly, there was no difference in thiamine levels among the eggs of known wild spawners (3,692 pmol/g) and eggs from redds of known wild spawners on Rodgers Creek (6,511 pmol/g) and eggs from redds of putative wild spawners on the Credit River (8,474 pmol/g). Thiamine levels of eggs from putative hatchery redds were significantly higher ($P < 0.05$) than eggs of known wild spawners (3,692 pmol/g) but not eggs from redds of putative wild spawners (8,474 pmol/g). The thiamine concentration of eggs from known wild redds on Rodgers Creek (6,511 pmol/g) differed ($P < 0.05$) from that of eggs from putative hatchery redds on the Credit River. In contrast, the eggs of putative wild redds (8,474 pmol/g) on the Credit River did not differ from that of putative hatchery redds ($P > 0.05$). The difference among groups did not appear related to the size of spawner since the thiamine concentration of eggs removed from

known wild spawners was unrelated ($r = 0.37$, $df = 5$, $P > 0.05$) to the size of spawner.

DISCUSSION

There were two groups apparent in the stable isotope data for redds possibly consistent with differing parentage, that being hatchery and wild. The lack of differences between known hatchery fish and redds of putative hatchery fish for both $\delta^{13}\text{C}$ (−17.7‰ versus −17.5‰) and $\delta^{15}\text{N}$ (9.9‰ versus 9.8‰) supports the designation of the latter group as being of hatchery origin. Similarly, the data for redds of presumed wild origin supports a designation of wild origin. Levels of $\delta^{13}\text{C}$ in eggs of putative wild redds (−26.6‰) were lower than that of eggs of known wild spawners (−22.2) and much lower than that of eggs from either known hatchery spawners (−17.7‰) or putative hatchery redds (−17.5‰), which we infer represent hatchery fish. In addition, levels of $\delta^{15}\text{N}$ in eggs from putative wild redds (16.4‰) were higher than that of eggs from known wild fish (15.0‰) and much higher than that of eggs from either known hatchery spawners (9.9‰) or putative hatchery redds (9.8‰).

As the stable isotope signatures of the two groups of eggs (e.g., wild and hatchery) are related to the diet of their maternal parents, the higher nitrogen values in putative wild Atlantic Salmon eggs suggests a longer food chain for their maternal parents than was the case for the maternal parents of hatchery-reared Atlantic Salmon eggs (Vander Zanden and Rasmussen 1999). It follows from this that the greater variation in the isotopic signatures of putative wild eggs compared to putative hatchery eggs probably represents greater heterogeneity in the diet of wild parents, possibly reflecting variation in the isotopic signatures of prey items (Kiriluk et al. 1995). By comparison, the diets of hatchery parents appeared quite homogenous probably due to the high level of regulation in feedstock materials used to formulate hatchery diets based on minimal variation in the isotopic signature of hatchery eggs. Dempson and Power (2004) found slight variation in isotopic signatures among various salmon aquaculture feeds, and such variation would be largely eliminated by the adoption of a standardized feed, as is the case for the OMNRFCS. More importantly however, hatchery diets use marine fish almost exclusively as the protein source (D. C. Honeyfield, personal communication) and it is known the $\delta^{13}\text{C}$ value is distinctly different between marine and freshwater–terrestrial animals. Signatures of marine animals are typically 7‰ more ^{13}C enriched relative to those from freshwater equivalents, and this difference was similar to that between putative hatchery and wild salmon eggs (i.e., 9‰; Fry et al. 1983). Given the near universal use of marine protein in hatchery diets, the separation observed in this study with Atlantic Salmon may be applicable to a broad range of species allowing the differentiation of hatchery broodfish and their eggs from that of wild adults and their eggs when diet consists of freshwater prey.

Although the use of stable isotope signatures of eggs offers considerable potential for differentiating the feeding origins of parents, little is known about eggs relative to that of other fish

tissues. Much of the literature pertaining to the use of stable isotopes for discerning feeding relationships among fish populations is based on muscle (Vander Zanden et al. 1997, 1999; Perga and Gerdeaux 2005) and liver (Perga and Gerdeaux 2005), with relatively little reported for eggs (Grey 2001; Acolas et al. 2008). Due to variation in tissue turnover rates, liver, because it is a regulatory tissue, appears to reflect annual changes in diet, whereas muscle, because isotope turnover depends primarily on growth rather than metabolic replacement, reflects food consumed during periods of growth (Perga and Gerdeaux 2005). As gonadosomatic growth depends on current food intake rather than tissues accumulated during past somatic growth (Roff 1983), it is reasonable to suppose that egg composition reflects diet. This is supported by the observation that for Atlantic Salmon, the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ concentration of the egg is highly correlated with that of muscle (Fitzsimons, unpublished data).

Within the wild Atlantic Salmon eggs obtained for this study, there was evidence of spatial variation in isotopic signatures between eggs from putative wild redds and known wild spawners, although the reasons for this variation are not clear. Significant differences in $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ were observed between eggs removed from known wild spawners collected 10 km from Lake Ontario and eggs removed from putative wild redds located approximately 40 km from the lake. The effect of diet on isotopic signature is considered to be greater for $\delta^{15}\text{N}$ than $\delta^{13}\text{C}$ (Peterson and Fry 1987), but the difference in $\delta^{15}\text{N}$ between the groups was less than one trophic level (e.g., 3.4‰; Vander Zanden and Rasmussen 1999). Although the diet of Atlantic Salmon while in Lake Ontario is unknown, potential Lake Ontario prey fish show variation for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ among prey species (Kiriluk et al. 1995), and depending on diet this may be reflected in egg stable isotope signatures.

The effect of maternal diet in influencing egg composition as revealed by the isotopic signature was also evident for egg thiamine. Although levels in wild groups varied, possibly a reflection of diet while in the lake (Fitzsimons and Brown 1998; Honeyfield et al. 2005), concentrations of thiamine in wild groups, regardless of whether the eggs came from redds or females, appear to represent physiologically adequate levels capable of preventing thiamine-dependent mortality. In all cases thiamine concentrations exceeded levels (1,600 pmol/g) considered protective of thiamine-dependent larval mortality (Ketola et al. 2000). Based on thiamine levels in Alewife-dependent Atlantic Salmon stocks reported by Fisher et al. (1998b) that averaged less than 1,000 pmol/g, the levels in the three groups of wild eggs examined here were three- to eight-fold higher. Although this would suggest a relatively low level of Alewife consumption by Lake Ontario Atlantic Salmon, additional work would be required to confirm this. Preliminary information based on the use of stable isotope mixing models indicates that over a 4-year period Alewives comprised no more than 10% of the diet of Lake Ontario Atlantic Salmon (Fitzsimons, unpublished data).

The relative abundance of wild to hatchery redds in our study, when combined with the known number of hatchery fish stocked, can be used to estimate both the reproductive efficiency (i.e., redds built per female) of hatchery Atlantic Salmon as well as the number of wild spawners present in the study area. Fleming et al. (1996) reported that Atlantic Salmon, whether wild or farm reared, construct one redd per female. In our study the proportion of hatchery to wild redds confirmed to have eggs was 74% based on 19 redds with assumed parental identity using stable isotopes. Therefore of the 79 redds identified, 58 redds would be attributable to hatchery females. The total number of mature hatchery females out of the 342 hatchery fish stocked was 162. Using this estimate of the number of female hatchery fish stocked and that spawned and built redds, it can be calculated there were 2.8 hatchery females per redd suggesting low reproductive efficiency. Although we do not know if some hatchery spawners may have migrated either farther upstream or downstream and not been censused in our surveys, which covered a restricted geographic area, this seems unlikely based on the distribution of redds observed. Our surveys of spawning activity were limited to 2,500 m of river, which included a 500-m stretch of river upstream and downstream of each of the areas where fish were released. Most redds observed were in the area where fish were released with far fewer redds found in the area immediately upstream and downstream of the release area suggesting that spawning occurred primarily in the area where fish were released. Alternatively, the hatchery fish used in this study, like their farmed counterparts (Fleming et al. 1996), may construct fewer redds per female than wild fish supporting the notion of low reproductive efficiency with over half the females not spawning. In an experimental study, Fleming et al. (1996) observed that wild Atlantic Salmon constructed almost two-fold more redds per female than hatchery-reared Atlantic Salmon. Of the 79 redds, we would estimate, based on proportions, that a total of 21 redds would be attributable to putative wild spawners, but since the number of wild adults was unknown we cannot estimate the number of wild spawners per redd. Based on one redd per female, at least 21 wild spawners may have spawned in the survey area, although if wild spawners performed similarly to hatchery females (e.g., 2.8 females/redd), this number may have been closer to 59 wild spawners.

Selection of spawning habitat that is inappropriate for reproduction can have profound fitness implications (Schjorring 2002) and affect survival (Chapman 1988). In this study though, we detected few measurable differences between the habitat used by hatchery and wild spawners. Of those differences, the apparent preference for riffle habitat over pool-riffle habitat by hatchery spawners may be because of the reduced upwelling flow in riffles as opposed to pool-riffles. This has potential to negatively affect embryonic survival, but more work would be required to confirm this. The wild spawners responsible for the wild redds identified in this study were likely the result of earlier stocking activities of hatchery-reared Atlantic Salmon so are not representative of a wild remnant stock. These Atlantic Salmon,

based on numbers and life stages of fish stocked, were probably of an earlier developmental stage than adults because of the much higher stocking of yearling and younger stages that has occurred (M. Daniels, Ontario Ministry of Natural Resources, personal communication). Nevertheless, despite the period of holding under hatchery conditions, the spawning habitat used by wild Atlantic Salmon in this study was similar to that given in reviews of the freshwater requirements of Atlantic Salmon based on wild fish (Gibson 1993).

Due to wide-scale reductions in native stocks and conservation needs (Carr et al. 2004), hatchery-reared Atlantic Salmon are often the only practical source of fish for restoring landlocked populations such as that in Lake Ontario, but they may be associated with reduced reproductive efficiency. Wild and stocked fish can breed together, but detrimental effects of stocking on native stocks have been reported, notably a fitness decline of wild populations (Youngson and Verspoor 1998; Jonsson and Jonsson 2006; Araki et al. 2007). In Lake Ontario, there has been no wild stock to draw on as the original stock was extirpated (COSEWIC 2006). As a result, restoration has been based entirely on the use of hatchery-reared fish. However, because of concerns regarding the reproductive efficiency of captive bred Atlantic Salmon from laboratory experiments (Fleming et al. 1996), assessing the proportion of wild fish during recovery and making corresponding adjustments to the stocking of hatchery fish may be important for maintaining a positive recovery trajectory. In our study, while spawning habitat selection appeared similar between hatchery and wild Atlantic Salmon, there was the suggestion that a relatively low proportion of hatchery-reared fish built redds. In addition, mortality of stocked fish is assumed to be elevated relative to their wild counterparts, given the relatively low percentages of released individuals returning to target rivers (Eckert 2003; Ruzzante et al. 2004; COSEWIC 2006). Stocked fish are also more likely to stray than wild conspecifics when returning to spawning grounds (Jonsson et al. 2003), and this may affect spawning habitat use. For Lake Ontario, it is recognized there is considerable heterogeneity in tributary habitat quality in the basin (Stanfield and Kilgour 2006; Randall 2010) and as a result the quality of spawning habitat probably varies as well (Stanfield and Jones 2003).

The complete loss of the historic Atlantic Salmon stock as occurred with Lake Ontario will make it more difficult, but seemingly not impossible, to establish a self-sustaining stock that initially, at least, relies almost entirely on hatchery-reared fish. Restoration of Atlantic Salmon in Lake Ontario is being pursued by the Ontario Ministry of Natural Resources through an adaptive management process involving research into a variety of rehabilitation measures, including habitat restoration and stocking at a variety of life stages. The timing, numbers, and appropriate life stages to use for this purpose are being evaluated on an experimental basis predicated in part on demographic analysis that suggests that the most effective way of improving population fitness is by increasing survival at early life stages (age-0, age-1, and age-2 smolts), but this is an ongoing process

(M. Koops, Fisheries and Oceans Canada, personal communication).

Although it is still uncertain as to the appropriate strategy to use for the restoration of Atlantic Salmon in Lake Ontario, the use of some life stages, like adults as used here, may have certain advantages over other life stages like juveniles. At low population size, being able to immediately access and potentially saturate available high-quality spawning habitat is critically important for achieving maximum reproductive rate (Myers et al. 1999). Restoration strategies employing stocking of mature hatchery-reared fish directly into spawning rivers can ensure that the best habitat in the best tributaries is used for spawning (Arahamian et al. 2003). Juveniles and younger stages can be released into streams believed to have superior habitat quality for spawning and rearing but can encounter mortality due to predation on their way downstream and in the lake (Klemetsen et al. 2003). Hence, the effectiveness of stocking of juveniles relative to the stocking of mature adults for rebuilding spawning stocks may be reduced. The effectiveness of juvenile stocking may be further reduced by straying prior to the initiation of the spawning migration (Jonsson et al. 2003). Those fish that successfully migrate to a high-quality stream may, in addition, encounter elevated stream temperatures before reaching the colder temperatures of the upper reaches of a stream, also resulting in mortality (Eriksen et al. 2006; Rand et al. 2006). Exposure to elevated temperatures can lead to mortality dependent on run timing, availability of groundwater, and ambient conditions (Everitt 2006; Farrell et al. 2008). It is important to note that Atlantic Salmon may be much more susceptible to the effects of climate-change-mediated increases in river water temperature (Eaton and Scheller 1996) than other salmonines. Atlantic Salmon, by entering spawning streams during summer months when highest water temperatures occur (Webb and McLay 1996), are likely to be exposed to much warmer temperatures than other salmonines, especially if coldwater pool habitat is limiting (Ketola et al. 2009).

Although there is considerable uncertainty as to the contribution that stocking at a given life stage can make towards the restoration of Atlantic Salmon in Lake Ontario, the technique elaborated here appears to be able to provide an objective means for evaluating reproductive effort by putative hatchery and wild fish. These in turn can provide a means of objectively evaluating relative success in achieving conservation spawner requirements (O'Connell et al. 1997).

This work represents the first documentation using naturally spawned eggs of natural reproduction by Atlantic Salmon in Lake Ontario, and the second time in the history of the program that natural reproduction has been observed. Natural reproduction was first documented by Johnson et al. (2010) from age-0 fish collected in the Salmon River (New York) in 2009. Taken together these observations represent an important milestone in the restoration program and, although a marked improvement over the previous decade (Crawford 2001), will require additional work towards linking the importance of individual restoration

activities to the establishment and relative abundance of spawning runs. Our findings follow a change in the Atlantic Salmon restoration program in 2006 to increase stocking to 2.5 million fry. These stocking efforts focused on three high-quality Lake Ontario tributaries, including the Credit River, suggesting the potential for an increased run size in the future (Marion Daniels, Ontario Ministry of Natural Resources, personal communication). Moreover, while it was evident from this study that spawning Atlantic Salmon were able to ascend a partial barrier located at Norval and access high-quality spawning habitat located on the upper Credit River, in the future upstream movement will be facilitated by a fishway constructed at this barrier in 2012 (Mark Heaton, Ontario Ministry of Natural Resources, personal communication).

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